

WI COOL DIGITAL STUDIO

Ubuntu Application Server Secure Baseline Project Report

wcds-app-01 | VMware Workstation Lab

Organisation	Wi Cool Digital Studio
Owner	Nawi Pero
System	wcds-app-01
Project period	4-5 August 2026
Report date	5 August 2026
Classification	Internal lab documentation

Project judgement

This configuration work is substantial enough to stand as an independent cloud-security portfolio project. It demonstrates operating-system administration, secure network configuration, patch governance, bootloader recovery, host firewalling, SSH hardening, validation, and recovery-point management.

No passwords, private keys, access tokens, or Azure subscription identifiers are included in this report.

Document control

Purpose	Record the work completed to prepare and harden the Ubuntu application server before the Azure cloud-security phase.
Primary audience	Lab owner, assessor, future administrator, or portfolio reviewer.
Evidence basis	Command outputs and screenshots captured during the guided configuration session.
Scope boundary	Ubuntu server baseline and supporting VMware/ADC1 dependencies. Azure tenant configuration is not included.

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1. Executive feedback

Assessment: portfolio-ready as a standalone infrastructure security project, with a small number of clearly documented follow-ups.

The work progressed well beyond routine VM setup. It established a repeatable security baseline, handled a complex operating-system upgrade, recovered from a GRUB bootloader configuration error, corrected a persistent network-definition mismatch, and closed with validated services and VMware recovery snapshots. The structured review-before-change approach reduced risk and produced useful evidence.

Area	Assessment	Feedback
Scope and complexity	Complete	The project covers compute, storage, identity naming, networking, patching, firewalling, SSH, service health and recovery.
Change safety	Strong	Package simulations, service validation and snapshots were used before disruptive changes.
Security improvement	Strong	Inbound access was reduced, SSH was hardened, updates were completed and failed services were cleared.
Operational evidence	Strong	Key outputs were captured for hostname, network, DNS, HTTPS, packages, kernel, firewall and services.
Project closure	Partial	A final reboot after the Netplan and wait-online changes was not completed before the hardened snapshot.
Automation maturity	Follow-up	The procedure was manual. A later Ansible or shell-based baseline would improve repeatability.

What was done particularly well

- Rollback gates were created before and after high-risk work.
- The Wazuh repository conflict was investigated rather than bypassed, preserving package-signing controls.
- The upgrade was simulated and the single removal was identified before approval.
- Wazuh Manager was deliberately held at version 4.12.0-1, keeping it separate for the next project.
- The accidental GRUB choice was recovered safely before rebooting.
- Firewall and SSH changes were validated before the session moved on.

Main improvement opportunities

- Perform one final reboot test after the Netplan and systemd wait-online changes.
- Test SSH remotely from LABSWIN-11-01 before disabling password authentication.
- Replace password-based SSH with managed keys, then set PasswordAuthentication no.
- Create a DHCP reservation for 10.1.10.52, or define an approved static server address.
- Review and disable unused listening services instead of relying only on UFW.
- Reduce the Ubuntu VM's 31 GiB RAM allocation to avoid starving other lab VMs; 8-12 GiB is a practical starting range when Splunk/Wazuh are not both active.

Proceeding recommendation

Proceed to Azure only after a brief closure check: reduce VM memory while powered off, power on the Ubuntu VM once, confirm ens37/DNS/default route, confirm zero failed units, then power it off again. This converts the baseline from 'configured and snapshotted' to 'reboot-validated'.

2. Project overview and objectives

Wi Cool Digital Studio is treated as a cloud-native start-up that will later demonstrate migration and hybrid-cloud patterns. This project prepared the local Ubuntu server so it can participate safely in that wider architecture.

Project title	Ubuntu Application Server Secure Baseline and Recovery
Primary asset	wcds-app-01
Platform	Ubuntu 24.04 LTS on VMware Workstation
Business owner	Nawi Pero
Supporting services	ADC1 for AD DS, DNS and DHCP; VMware VMnet8 for NAT egress
Success criterion	A patched, named, network-stable and hardened Ubuntu server with validated recovery points and no failed systemd units.

Objectives

- Establish a reliable inventory of compute, memory, storage and network configuration.
- Rename the server using an organisation-specific convention.
- Repair package repository trust and complete the Ubuntu upgrade without changing Wazuh.
- Ensure GRUB and the upgraded kernel boot successfully.
- Restrict inbound access with UFW and harden OpenSSH.
- Enable automatic security updates and clear failed service states.
- Correct persistent Netplan configuration to use the real Wi Cool interface.
- Create recoverable VMware snapshots at major change gates.

Out of scope

- Azure tenant, subscription, landing-zone and identity configuration.
- Wazuh Manager activation, tuning or SIEM use cases.
- Splunk configuration or data onboarding.
- Application deployment and application-specific security controls.
- Penetration testing from Kali.

3. Environment and architecture

Wi Cool Digital Studio - VMware Lab Topology

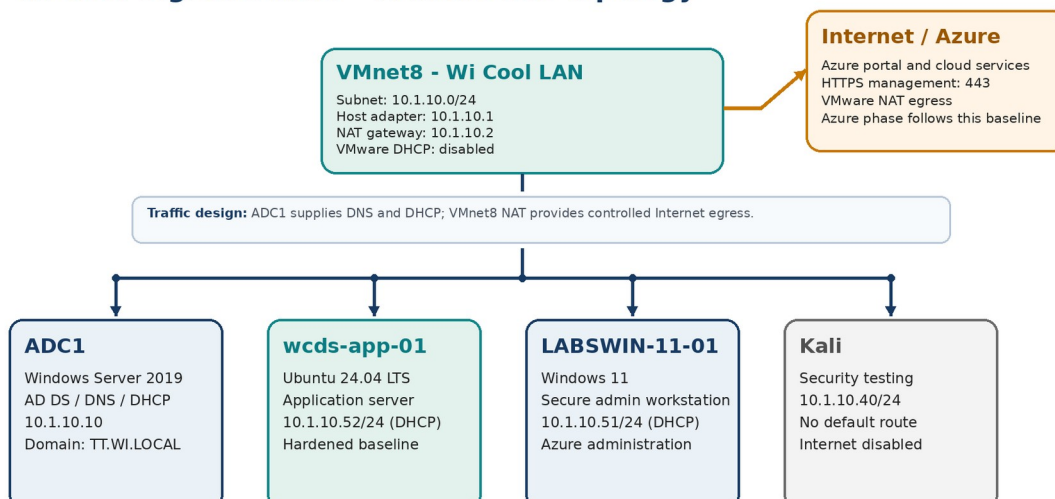


Figure 1. Wi Cool Digital Studio VMware lab topology.

Environment inventory

Component	Role	Key configuration
ADC1	Legacy identity and infrastructure	Windows Server 2019; TT.WI.LOCAL; 10.1.10.10; AD DS, DNS and DHCP.
wcds-app-01	Ubuntu application server	Ubuntu 24.04 LTS; 10.1.10.52/24; ens37; hardened in this project.
LABSWIN-11-01	Secure administration workstation	Windows 11; 10.1.10.51/24; domain-connected; used for Azure administration.
Kali	Security testing	10.1.10.40/24; no default route; internet disabled by default.
VMnet8	Internal/NAT network	10.1.10.0/24; host 10.1.10.1; gateway 10.1.10.2; VMware DHCP disabled.

Server resource baseline

CPU	4 vCPU, x86-64, VMware full virtualisation; Intel Core i5-10210U host CPU exposed.
Memory	31 GiB RAM; 2.4 GiB used and approximately 28 GiB available at measurement time; 4 GiB swap unused.
Virtual disk	500 GB.
Boot partition	2 GB ext4 mounted at /boot.
LVM physical volume	Approximately 498 GB.
Root logical volume	Approximately 100 GB ext4; 98 GB usable; 27 GB used; 67 GB available; 29% utilised.
Free LVM capacity	Approximately 398 GB retained for later workload-specific allocation.

Network baseline

Hostname	wcds-app-01
Primary interface	ens37
IPv4 address	10.1.10.52/24 via DHCP
Default gateway	10.1.10.2 via ens37
DNS server	10.1.10.10 (ADC1)
DNS search domain	TT.WI.LOCAL
Netplan renderer	NetworkManager
Cloud-init networking	Disabled to prevent regeneration of the stale ens33 configuration.

4. Baseline implementation

4.1 Host identity and local resolution

The default hostname `osboxes` was replaced with `wcds-app-01`. The local hosts mapping was updated from `127.0.1.1 osboxes` to `127.0.1.1 wcds-app-01`. Static hostname, local resolution and hostname persistence were validated.

4.2 Repository trust and package maintenance

APT initially failed because the Wazuh repository appeared three times: two signed entries and one unsigned entry. The source file was reduced to a single entry using `/usr/share/keyrings/wazuh.gpg`. Package catalogue refresh then completed successfully.

```
deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main
```

The full-upgrade simulation proposed 560 upgrades, 10 new packages and one removal. The removal of `libglapi-mesa` was reviewed as part of the Mesa graphics-package transition. Wazuh Manager was placed on hold, producing a final plan of 559 upgrades, 10 new packages, one approved removal and one held package.

4.3 Wazuh preservation

Package	wazuh-manager
Installed version	4.12.0-1
APT state	Held
Runtime state	Inactive by project decision
Reason	Reserved for a separate Wazuh project and protected from unintended upgrade/removal.

4.4 GRUB and kernel recovery

During the operating-system upgrade, the GRUB package reported that its previous installation target was no longer present or had changed. No target was selected and 'continue without installing GRUB' was accepted accidentally. The VM was not rebooted. GRUB was then installed explicitly to the whole virtual disk `/dev/sda`, and the configuration was regenerated.

```
sudo grub-install /dev/sda
sudo update-grub
sudo dpkg --configure -a
sudo apt-get check
```

The regenerated menu detected kernel `6.8.0-137-generic` and retained older kernels `6.8.0-63-generic` and `6.8.0-58-generic` as recovery options. A subsequent reboot succeeded and `uname -r` confirmed `6.8.0-137-generic`.

4.5 Host firewall

UFW was already active with a default deny-incoming policy, but SSH was allowed from anywhere over IPv4 and IPv6. A trusted-network rule was added first, then the unrestricted rules were removed.

```
Default: deny (incoming), allow (outgoing), disabled (routed)
22/tcp ALLOW IN 10.1.10.0/24
Logging: on (low)
```

4.6 OpenSSH hardening

A dedicated configuration fragment was created at `/etc/ssh/sshd_config.d/10-wicool-hardening.conf`. The configuration was syntax-tested before reloading the SSH service.

```
PermitRootLogin no
MaxAuthTries 4
X11Forwarding no
```

Effective settings after reload: root login disabled, maximum attempts four, X11 forwarding disabled, public-key authentication enabled and password authentication temporarily enabled.

4.7 Automatic updates and service health

unattended-upgrades was confirmed enabled and active. A systemd health check identified systemd-networkd-wait-online.service as failed. The failure was investigated rather than ignored.

4.8 Netplan correction

The persistent cloud-init Netplan file still defined ens33 even though ens37 was the active Wi Cool interface. Cloud-init network generation was disabled, the original file was backed up, and Netplan was rewritten to use NetworkManager with DHCP on ens37.

```
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    ens37:
      dhcp4: true
```

netplan generate and netplan apply completed without errors. ens37 retained 10.1.10.52/24. Because NetworkManager-wait-online.service was already enabled, the redundant systemd-networkd-wait-online.service was disabled and its stored failed state was reset.

5. Validation and acceptance evidence

Control or test	Observed evidence	Result
Hostname persistence	hostnamectl --static returned wcds-app-01 after reboot.	Passed
Local hostname resolution	getent hosts wcds-app-01 resolved to 127.0.1.1.	Passed
Internal DNS	ADC1.TT.WI.LOCAL resolved to 10.1.10.10.	Passed
IPv4 interface	ens37 UP with 10.1.10.52/24.	Passed
Default route	default via 10.1.10.2 dev ens37.	Passed
DNS service	resolvectl showed 10.1.10.10 and TT.WI.LOCAL.	Passed
Azure HTTPS	curl -I https://portal.azure.com returned HTTP/2 403, proving DNS, routing and TLS reachability.	Passed
Kernel boot	uname -r returned 6.8.0-137-generic.	Passed
APT integrity	apt-get check completed with no dependency errors.	Passed
Upgrade completeness	0 upgraded, 0 newly installed, 0 to remove and 1 held.	Passed
Wazuh preservation	wazuh-manager 4.12.0-1 remained installed and held.	Passed
UFW policy	Only 22/tcp from 10.1.10.0/24 is allowed	Passed

Control or test	Observed evidence	Result
	inbound.	
SSH health	systemctl is-active ssh returned active.	Passed
Automatic updates	unattended-upgrades enabled and active.	Passed
Failed services	systemctl --failed reported 0 loaded units.	Passed
Final reboot after Netplan change	Not performed before the hardened snapshot.	Follow-up

Acceptance statement

The server met the planned technical baseline at the time of shutdown and snapshot creation. The only closure qualification is the absence of a final reboot after the last Netplan and systemd service changes.

6. Problems encountered and recovery actions

Issue	Impact	Resolution	Lesson
Conflicting Wazuh Signed-By values	APT could not read source lists.	Removed duplicate and unsigned entries; retained one signed repository entry.	Repository trust errors should be corrected, not bypassed.
Large pending upgrade	560 packages required review.	Simulated the transaction, identified the removal and held Wazuh before approval.	Use dry runs and explicit package holds for change control.
GRUB target not selected	Potential failure to boot after kernel update.	Installed GRUB to /dev/sda, regenerated grub.cfg, checked packages and reboot-tested.	Never reboot after a bootloader warning until the boot path is repaired and validated.
Stale ens33 Netplan definition	Boot wait-online timeout and future network inconsistency.	Disabled cloud-init networking, backed up the file and defined ens37 under NetworkManager.	Persistent configuration must match the actual interface, not merely the current runtime state.
systemd wait-online failure	One failed unit remained after boot.	Confirmed NetworkManager ownership, disabled redundant networkd wait service and reset the failure state.	Remove service-manager overlap after confirming the intended renderer.
Host RAM pressure when starting Windows later	VMware could not initially power the Windows VM.	Not yet remediated; reduce powered-off VM allocations before concurrent use.	Resource right-sizing is part of secure and reliable lab operations.

7. Security assessment and NIST alignment

This section is an alignment assessment, not a claim of formal NIST compliance. The lab demonstrates selected outcomes and principles.

NIST CSF 2.0 current-profile view

CSF Function	Evidence from this project	Assessment
GOVERN	Named owner, defined scope, change gates, package hold decision and documented residual risks.	Partial
IDENTIFY	Asset, network, storage, service and listening-port inventory captured.	Achieved

CSF Function	Evidence from this project	Assessment
PROTECT	Patching, repository signing, firewall restriction, SSH hardening, automatic updates and configuration backup.	Achieved
DETECT	Listening services and failed units were reviewed; Wazuh/Splunk monitoring was not activated as part of this project.	Partial
RESPOND	The GRUB incident was contained, diagnosed and recovered without rebooting into an uncertain state.	Partial
RECOVER	Multiple VMware snapshots and older kernels provide rollback paths; a full snapshot restore test remains outstanding.	Partial

NIST CSF 2.0 states that its Functions should be addressed concurrently and that Current and Target Profiles can be used to document posture and prioritise gaps [1]. This report therefore serves as a small Current Profile and improvement register for wcds-app-01.

Zero Trust interpretation

Restricting SSH to 10.1.10.0/24 is useful network segmentation, but it is not a complete zero trust control. NIST SP 800-207 emphasises that trust should not be granted solely because a user or device is on an internal network; access should be least-privilege, resource-focused and continually evaluated [2]. The Azure phase should add identity-based access, MFA, device posture and per-resource authorisation.

NIST SP 800-215 similarly highlights endpoint management, patch status, authentication and contextual information for secure hybrid access [3]. The patching, endpoint inventory and hardened administration performed here are foundational steps, while stronger identity and device-health enforcement remain future work.

Control strengths

- Default-deny inbound host firewall.
- Reduced SSH exposure and disabled root login.
- Signed package repository and completed security updates.
- Recovery snapshots and retained older kernels.
- Validated DNS, routing and TLS reachability.
- Separation of the Wazuh project through a deliberate package hold.

Control limitations

- SSH password authentication remains enabled.
- Subnet membership is still used as the main SSH trust boundary.
- No central identity, MFA or device-compliance decision is applied to the server.
- Several unused services continue listening locally or on all interfaces, even though UFW blocks inbound access.
- The server uses DHCP without a documented reservation.
- No vulnerability scan, configuration benchmark scan or restore test was completed.

8. Residual risks and recommendations

Priority	Recommendation	Reason / acceptance criterion
P1	Complete a final reboot validation.	After reboot, ens37 must receive 10.1.10.52 or the approved reservation; gateway, DNS and systemctl --failed must pass.
P1	Right-size VM memory.	Reduce from 31 GiB while powered off. Start with 8-12 GiB and monitor when Splunk or Wazuh is active.

Priority	Recommendation	Reason / acceptance criterion
P1	Establish an address reservation.	Reserve 10.1.10.52 for the Ubuntu VM in Windows DHCP, or approve a static address outside the dynamic pool.
P2	Configure SSH keys from LABSWIN-11-01.	Test key login, then disable PasswordAuthentication and retain console recovery access.
P2	Review exposed listeners.	Disable XRDP, GNOME remote desktop, Avahi, Splunk or Mongo-related services when not required.
P2	Test snapshot recovery.	Restore a clone or isolated copy from the hardened snapshot and confirm boot and network health.
P3	Automate the baseline.	Convert the repeatable controls to Ansible or version-controlled shell scripts with verification steps.
P3	Add security assessment tooling.	Run a benchmark or vulnerability scan and record findings in a separate remediation plan.
P3	Activate monitoring in its own project.	Unhold, update and configure Wazuh only when the next project begins and rollback evidence exists.

Suggested project artefacts for a portfolio

- This PDF report.
- A redacted screenshot set showing the key validation outputs.
- A one-page architecture diagram.
- A change log with timestamps and snapshot names.
- A short lessons-learned statement covering the GRUB recovery and Netplan mismatch.
- A future automation repository containing the hardened baseline.

9. Project conclusion

The Ubuntu baseline project successfully transformed a general-purpose VMware guest into a documented application-server foundation. The server was inventoried, renamed, patched, recovered from a bootloader risk, placed behind a default-deny host firewall, hardened for SSH, corrected to use the intended network interface, and left with no failed systemd units. Wazuh was preserved for the next project rather than mixed into this scope.

Final decision

Treat this as Project 1: 'Ubuntu Application Server Secure Baseline and Recovery'. Treat the Azure landing zone as Project 2. Complete the short closure checks in Section 8, then proceed.

Appendix A. Key commands and configuration

Inventory and validation

```
hostnamectl --static
lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINTS
sudo vgs
df -hT /
lscpu
ip -br address show ens37
ip route show default
resolvectl dns ens37
resolvectl domain ens37
getent hosts ADC1.TT.WI.LOCAL
curl -I https://portal.azure.com
systemctl --failed --no-pager
```

Package and GRUB recovery

```
sudo apt update
sudo apt-get --simulate full-upgrade
sudo apt-mark hold wazuh-manager
sudo apt-get full-upgrade
sudo grub-install /dev/sda
sudo update-grub
sudo dpkg --configure -a
sudo apt-get check
uname -r
```

Firewall and SSH

```
sudo ufw allow from 10.1.10.0/24 to any port 22 proto tcp
sudo ufw status numbered
sudo ufw status verbose
sudo sshd -t
sudo systemctl reload ssh
sudo sshd -T
```

Netplan and service correction

```
echo 'network: {config: disabled}' | sudo tee /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg
sudo cp /etc/netplan/50-cloud-init.yaml /etc/netplan/50-cloud-init.yaml.bak
sudo netplan generate
sudo netplan apply
sudo systemctl disable --now systemd-networkd-wait-online.service
sudo systemctl reset-failed systemd-networkd-wait-online.service
```

Important operational note

Commands are recorded for evidence and learning. They should not be copied into another system without first confirming interface names, boot mode, package versions, service ownership and recovery options.

Appendix B. Snapshot register

Snapshot	Purpose	Status
03-Before-Ubuntu-Password-Reset-2026-08-04	Existing recovery point created when password recovery was considered; reset was not needed.	Retained
04-Before-Ubuntu-Full-Upgrade-2026-08-04	Powered-off rollback point before the large package and kernel upgrade.	Created
05-After-Ubuntu-Upgrade-2026-08-05	Powered-off recovery point after GRUB repair, kernel boot and connectivity validation.	Created
06-Ubuntu-Hardened-Baseline-2026-08-05	Final powered-off snapshot after firewall, SSH, Netplan and service-health work.	Created

References

- [1] National Institute of Standards and Technology (2024), *The NIST Cybersecurity Framework (CSF) 2.0*, NIST CSWP 29, <https://doi.org/10.6028/NIST.CSWP.29>.
- [2] Rose, S.; Borchert, O.; Mitchell, S.; Connelly, S. (2020), *Zero Trust Architecture*, NIST SP 800-207, <https://doi.org/10.6028/NIST.SP.800-207>.
- [3] Chandramouli, R. (2022), *Guide to a Secure Enterprise Network Landscape*, NIST SP 800-215, <https://doi.org/10.6028/NIST.SP.800-215>.

End of report